Fiscal Year:	FY 2023	**Task Last Updated:**	FY 04/29/2024
PI Name:	Zawieja, David Ph.D.		
Project Title:	Effects of Microgravity on Ocular Vascular Hydrodynamics		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	dcz@tamu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	979-436-0829
Organization Name:	Texas A&M University		
PI Address 1:	Department of Medical Physiology		
PI Address 2:	8447 Riverside Parkway		
PI Web Page:			
City:	Bryan	**State:**	TX
Zip Code:	77807	**Congressional District:**	31
Comments:			
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Start Date:	12/21/2018	**End Date:**	12/20/2022
No. of Post Docs:	6	**No. of PhD Degrees:**	30
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	11
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/20/2022 (original end date was 12/20/2021) per NSSC information (Ed., 1/4/22)		
Key Personnel Changes/Previous PI:	2023 Update: As Principal Investigator (David Zawieja, Ph.D.) is retiring from Texas A&M University in a few years, the new PI for this project will be Travis Hein, Ph.D. (also at Texas A&M University). Dr. Zawieja will remain as a CoInvestigator on the project. 2022 Update: Former PI (Dr. Anatoliy Gashev M.D., Ph.D.) unexpectedly passed away in August 2021. His technician still worked on this project. 2020 update, Dr. Binu Tharakan moved from Scott and White Memorial Hospital in Temple TX to Morehouse School of Medicine in Atlanta, Georgia.		
COI Name (Institution):	Loerch, Linda M.S. (NASA Johnson Space Center) Tharakan, Binu Ph.D. (Scott & White Memorial Hospital) Macias, Brandon Ph.D. (NASA Johnson Space Center) Lee, Stuart Ph.D. (Wyle Laboratories, Inc./NASA Johnson Space Center) Hein, Travis Ph.D. (Texas A&M University System) Bagher, Pooneh Ph.D. (Texas A&M University System)		
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Task Description:

Spaceflight-associated neuro-ocular syndrome (SANS) develops in astronauts completing long duration spaceflights. It is reported to affect ~70% of astronauts completing long-duration spaceflights (NASA Evidence Report 4/2022) and has been characterized as the development of one or more findings: optic disc edema, hyperopic shifts, globe flattening, cotton-wool spots, or choroidal folds. These changes in ocular function could lead to vision alteration that is uncorrectable and mission threatening. The leading hypothesis for the development of ocular changes is that prolonged exposure to the headward fluid shift that occurs in weightlessness is the primary instigating factor, and additional factors such as genetic disposition, anatomic variations, ambient CO2 on the International Space Station, or on-orbit exercise countermeasures may augment or diminish the development of ocular symptoms. However, the pathophysiology of SANS remains unclear. The changes in fluid pressures associated with the loss of the gravitational vector and the resulting headward fluid shifts can influence many ocular-associated fluid compartments, CSF in the brain and optic nerve, blood and lymph vessels in the brain, retina and eye. Previous work in the brain microcirculation found alteration in the regulation of their function. However studies of the vessels in the eye have not previously been done. Therefore, studies of ocular vascular hydrodynamics are required to clarify if chronic mild elevations of ocular pressure variables compromise ocular structure and function. Since all blood and lymph vessels are compliant, fluid-filled structures whose pressures are strongly influenced by gravity, we focused the RR23 studies on the potential changes directly to the ocular vasculature caused by microgravity. Perfusion of the optic nerve and inner retina for sufficient delivery of oxygen and nutrients is dependent on retinal blood flow. The pressure gradient for driving blood flow through the inner retina begins with the arterial pressure in the feed artery, which is the central retinal artery in humans. Changes in retinal blood flow or pressure may contribute to the formation of cotton wool spots and optic disc edema. Optic disc edema, choroidal folds, and optic nerve thickening may also result from ocular venous congestion and/or elevated venous compliance, disruption of the blood-retinal barrier, and/or reduction in ocular lymph flow. There had been no systematic analysis of the ocular vascular changes in microgravity. We assembled a team of experts in SANS and all 3 main vascular types (arteries, veins, and lymphatics) to address this information gap. Thus, the primary objective of this application is to determine whether microgravity alters the structure and function of the ocular vasculature at the level of feed arteries, venous exchange and capacitance vessels, and lymph vessels. This provides a novel comprehensive evaluation of the ocular vascular elements after spaceflight. The central hypothesis of this proposal is that microgravity/spaceflight-induced changes in the structure/function of the ocular vasculature lead to alterations in ocular hydrodynamics and promoted symptoms of SANS. We accomplished this objective using in vivo measures of ocular characteristics, ocular vascular function (retinal artery blood flow, retinal arteriole and venular diameter measurements, and retinal venular permeability measures) and in vitro studies of freshly isolated vascular structure and function (vessel/tissue histology, arterial vasomotor regulation, venous compliance measures, and lymphatic transport characteristics). These studies were conducted in mice flown in space for ~37 days immediately upon arrival back on earth (10.5 hours after splashdown) and the corresponding ground controls to address the following specific aims:
1: Evaluate the effects of microgravity on ocular artery structure/function. • In vivo Measurements (anesthetized mice): • VisualSonics Vevo 2100 ultrasound system was used for blood velocity measurement. The central retinal artery was located by color Doppler. • A TonoLab tonometer was used to measure intraocular pressure (IOP). • The Heidelberg Spectralis HRA + OCT system was used to acquire fundus and cross-sectional images of the mouse retina. • In vitro Functional Studies of the Ophthalmic Artery: • Isolated ophthalmic arteries were cannulated with glass micropipettes and pressurized to 75 cmH2O. Vessel diameter changes/contractile status to acetylcholine, sodium nitroprusside, and endothelin-1 were monitored and recorded.

2: Evaluate the effects of microgravity on ocular vein structure/function. • In vitro Functional Studies of the Angular Vein: • The mouse angular vein, which drains the supratrochlear and supraorbital vein, was isolated and mounted in a wire myograph. • Assessment of Retinal Microvascular Barrier Function: • Intravital microscopy (multiphoton laser scanning microscope) was used to acquire real-time images of FITC-dextran extravasation in anesthetized mice to measure microvascular permeability in the mouse retina. • Mice were injected with FITC-dextran (10 kDa) via the jugular vein, followed by intravital imaging of the retinal microcirculation every 5 min for 40 min. Retinal microvascular permeability/barrier integrity was measured based on extravasation fluorescence intensity of FITC-dextran.

3: Evaluate the effects of microgravity on ocular lymphatic structure/function. • Assessment of Ocular Lymphatic Vessel Function: • Isolated ocular collecting lymphatics were cannulated with glass micropipettes and pressurized from 0.5 to 4 cmH2O. Diameter changes were tracked and recorded to measure phasic and tonic contractile/pump function. • Assessment of Ocular Lymphatic Tissue Structure: • Perilymphatic tissue was collected for immunohistochemical analysis of local immune cells (mast cells, macrophages, dendritic cells, and lymphocytes) and for multiplex analysis of cytokines.

In addition to conducting experiments to address these primary missions we also0used a team of scientists to conduct the dissection and specimen storage for NASA's Biospecimen Sharing Program, since this occurred during the COVID pandemic and NASA did not allow their normal BSP team top travel. As a result of our BSP efforts we were granted permission to acquire some additional tissues for study.

A brief summary of the results from these novel studies (as seen below) provided the first comprehensive analysis of the effects of microgravity on ocular vascular function where the predominant changes associated with SANS in astronauts occur. These results will help define the roles the ocular vascular function may play in the etiology of SANS and could lead to the development of countermeasures for SANS.

• Aim 1: Effects of microgravity on ocular artery structure/function • Central retinal artery blood flow velocity was lower in the flight mice. • Trends towards higher IOP in the flight mice. • Trends towards increased total retinal thickness in the flight mice. • Ophthalmic arteries from flight mice exhibited normal vasodilator function but diminished basal vessel tone and endothelin-1-induced vasoconstriction.

• Aim 2: Effects of microgravity on ocular vein structure/function • Increase in retinal microvascular permeability. • The trends observed in isolated angular veins were attenuated vasoconstriction and vasodilation to adrenergic stimulus and exogenous nitric oxide donors, respectively.

• Aim 3: Effects of microgravity on ocular lymphatic structure/function • There was a large decrease (~40%) in ocular lymphatics from flight that had pumping activity. • Trends towards decreased lymph pump strength and increased pump frequency in those that exhibited intrinsic pumping activity. • Complete loss of shear-dependent dilation and trends towards impaired shear-dependent reductions in pumping

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The expected findings from the experimental planned for RR23 will also provide new insight into vascular complications relevant to ocular diseases in humans on Earth, such as glaucoma, diabetic macular edema, and ocular hypertension.
Task Progress:	Spaceflight Associated Neuro-ocular Syndrome (SANS) develops in astronauts completing long-duration spaceflights and is diagnosed based on one or more findings: optic disc edema, hyperopic shifts, globe flattening, or choroidal folds. Prolonged exposure to the headward fluid shift that occurs in weightlessness is regarded as the primary instigating factor for ocular changes, but the pathophysiology of SANS remains unclear. Also, there has been no systematic analysis of the ocular vascular changes in microgravity. The central hypothesis of this project is that microgravity/spaceflight-induced changes in the structure/function of the ocular vasculature, at the level of arteries, veins, and lymphatics, lead to alterations in ocular hydrodynamics and promote signs of SANS. This hypothesis was tested with the following Specific Aims: Specific Aim 1: Evaluate the effects of microgravity on ocular artery structure/function. Aim 1A: We completed the in situ studies by performing intraocular pressure (IOP) measurements, optical coherence tomography (OCT) /retinal fundus imaging, and Doppler ultrasound measurements in the flight, habitat ground control (HGC), and vivarium ground control (VGC) cohorts of mice. We found that central retinal artery blood flow velocity was lower in the flight mice compared to the HGC and VGC mice. IOP was about 20% higher in the spaceflight mice than in the VGC mice. Data analysis of the OCT images showed trends toward increased total retinal thickness in the flight mice. Aim 1B: We completed the in vitro studies for assessment of vasomotor function of isolated and pressurized ophthalmic arteries in all 3 cohorts of mice. We found that the dilations of isolated ophthalmic arteries to endothelium-dependent agonist acetylcholine and endothelium-independent agent sodium nitroprusside were not significantly different among cohorts. By contrast, basal tone and constriction to endothelin-1 were lower in ophthalmic arteries from flight mice. We collected the eyes and plasma from all 3 cohorts of mice and attempted to perform assays to determine the cytokine/inflammatory protein levels in these samples. Specific Aim 2: Evaluate the effects of microgravity on ocular vein structure/function. Aim 2A: Following the mission to the International Space Station (ISS), vasomotor responses were measured in isolated angular veins from nine C57BL/6J mice using wire myography. The angular vein was mounted on the jaw of a Danish Myotech wire myograph on 15µm gold-plated tungsten wire. Following a normalization procedure, veins were exposed to agonists that allowed for the examination of endothelial cell and smooth muscle cell function. Vivarium controls – consisting of mice of the same cohort housed in conventional vivarium cages; and habitat controls – consisting of mice housed in specially designed habitats exposed to similar CO2 levels, temperatures, etc., as on the ISS; were used for comparison. Preliminary analysis demonstrates that vasoconstriction to the adrenergic receptor agonist was reduced following spaceflight as compared to vivarium controls, with ground controls demonstrating an intermediate response. Vasodilation to an exogenous nitric oxide donor was reduced following spaceflight as compared to both vivarium and ground controls. These data suggest that angular vein function is altered following exposure to the space environment. Aim 2B: We performed experiments to assess the function of ocular veins after 37 days of spaceflight (RR23) in male mice. Retinal microvascular permeability was assessed in vivarium, ground control, and space mice. The mice were anesthetized with isoflurane and injected (iv) with FITC-dextran-10kDa as an indicator of vascular permeability. This was followed by the imaging of the retinal venules for FITC-dextran extravasation under a multi-photon microscope. We observed significant increases in venular permeability in the flight group compared to the vivarium and ground control groups after FITC-dextran injection. The ocular tissue from another set of mice from each group were collected and processed for immunohistochemistry and molecular biology studies. The work demonstrated trends towards reductions in the expression of the blood-retinal barrier (BRB) tight junction proteins (zonula occludens-1, claudin-5, and occludin). Specific Aim 3: Aim 3A: Evaluate the effects of microgravity on isolated ocular lymphatic structure/function. ~40% of the ocular lymphatics in the flight mice had no phasic pumping activity, whereas ~80% of all of the control groups did. There was no significant change in the resting tone of the lymphatics in flight versus controls, but there was a complete loss of the shear-induced inhibition of tone seen in normal/control ocular lymphatics. There were trends toward: decreased pump amplitude, increased lymph pump contraction frequency, and impaired flow/shear-dependent impact on the lymph pump amplitude and flow in flight versus controls. Aim 3B: In a subset of the mice (n= 5 side per group) we performed ocular lymph flow tracer studies by injecting trace amounts of a large molecular weight dextran into the tissues of both eyes (left and right) associated with the primary draining ocular lymphatics followed by acquisition and tracer analysis of both of the superficial cervical lymph node 15 minutes after injection to get an assessment of lymph transport in the 3 cohort groups. Ocular lymph transport was greatly decreased (> 300%) in the flight group compared to either control or the lumped controls (that were not statistically different from each other). This direct measure of ocular lymph transport shows significant declines in ocular lymph flow from the flight mice compared to both control groups and provides a direct correlation to the impaired lymphatic pump function observed in isolated ocular lymphatics from the flight group.
Bibliography Type:	Description: (Last Updated: 04/24/2019)